Aurora: Module Relay x8

Categories

Prepared for

Technical Staff

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Disclosure is prohibited

Revision History

Version	Date	Author	Description
0.1.0	2026-07-02	Martin Pellizzer	Starting version

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1. CORE

1.1 CORE Communicate

- The Relay Module shall communicate to CORE

1.1.1 The Relay Module shall communicate to CORE

Statements

- The Relay Module shall communicate to CORE via Modbus RTU protocol
- The Relay Module shall communicate to CORE via RS-485 hardware standard

1.2 CORE Execute

- The Relay Module shall execute relay-output commands (ON/OFF) received from CORE
- The Relay Module shall execute relay-output commands (ON/OFF) deterministically
- The Relay Module shall execute relay-output commands (ON/OFF) within specified time
- The Relay Module shall execute normal operation only when explicit reauthorization from CORE is given

1.2.1 The Relay Module shall execute relay-output commands (ON/OFF) received from CORE

Statements

- The Relay Module shall execute relay-output commands received from CORE
- The Relay Module shall manage the eight relay outputs to control or influence the ozone-generation process by operating the required process equipment and auxiliaries

1.2.2 The Relay Module shall execute relay-output commands (ON/OFF) deterministically

Statements

- The Relay Module shall execute commands deterministically

1.2.3 The Relay Module shall execute relay-output commands (ON/OFF) within specified time

Statements

- The Relay Module shall turn ON/OFF outputs within specified time

1.2.4 The Relay Module shall execute normal operation only when explicit reauthorization from CORE is given

Statements

- The Relay Module shall require explicit reauthorization from CORE before returning to normal operation

1.3 CORE Report

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- The Relay Module shall report to CORE outputs status
- The Relay Module shall report to CORE communication status
- The Relay Module shall report to CORE diagnostics
- The Relay Module shall report to CORE diagnostics identification
- The Relay Module shall report to CORE fault information
- The Relay Module shall report to CORE individual channel faults
- The Relay Module shall report to CORE any condition preventing reliable output control
- The Relay Module shall report to CORE module health
- The Relay Module shall report to CORE electrical identification
- The Relay Module shall report to CORE thermal identification
- The Relay Module shall report to CORE firmware identification
- The Relay Module shall report to CORE hardware identification

1.3.1 The Relay Module shall report to CORE its status

Statements

- The Relay Module shall report its status to CORE

1.3.2 The Relay Module shall report to CORE outputs status

Statements

- The Relay Module shall provide CORE with output status
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- The Relay Module shall report the state of all eight outputs

1.3.3 The Relay Module shall report to CORE communication status

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1.3.4 The Relay Module shall report to CORE diagnostics

Statements

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1.3.5 The Relay Module shall report to CORE diagnostics identification

Statements

- The Relay Module shall report diagnostics identification

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Statements

- The Relay Module shall provide sufficient fault information to CORE when in failure state
- The Relay Module shall report its faults to CORE
- The Relay Module shall report the fault as required when it detects an internal fault capable of compromising safe operation

1.3.7 The Relay Module shall report to CORE individual channel faults

Statements

- The Relay Module shall report individual channel faults

1.3.8 The Relay Module shall report to CORE any condition preventing reliable output control

Statements

- The Relay Module shall report any condition preventing reliable output control

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1.3.13 The Relay Module shall report to CORE hardware identification

Statements

- The Relay Module shall report hardware identification

1.4 CORE Monitor

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- The Relay Module shall monitor timeout with CORE to transition to its outputs to predefined safe state

1.4.1 The Relay Module shall monitor communication timeout with CORE

Statements

- The Relay Module shall detect communication timeout with CORE

1.4.2 The Relay Module shall monitor timeout with CORE to transition to its outputs to predefined safe state

Statements

- The Relay Module shall transition to its outputs to predefined safe state when timeout with CORE
- The Relay Module shall autonomously transition every output to its predefined safe state when communication with the CORE is lost beyond the defined safety timeout
- The Relay Module shall maintain a predefined safe state when communication with the CORE is lost beyond the defined safety timeout

2. Module

2.1 Module Relay Driver

- The Relay Module shall provide eight indipendently controlled relay-driven channels
- The Relay Module shall manage activation of each driver channel
- The Relay Module shall manage deactivation of each driver channel
- The Relay Module shall energize each relay coil independently across the full specified supply
- The Relay Module shall energize its assigned relay coil across the full specified temperature tolerance range
- The Relay Module shall energize its assigned relay coil with adequate voltage margin
- The Relay Module shall energize its assigned relay coil with adequate current margin
- The Relay Module shall energize its assigned relay coil with continuous-duty capability
- The Relay Module shall energize its assigned relay coil with short-circuit protection
- The Relay Module shall energize its assigned relay coil with overload protection
- The Relay Module shall energize its assigned relay coil with no unintended activation during reset conditions
- The Relay Module shall energize its assigned relay coil with no unintended activation during fault conditions
- The Relay Module shall limit the coil-generated transient to a safe level without exceeding driver ratings
- The Relay Module shall limit the coil-generated transient to a safe level without causing unacceptable relay release-time delays
- The Relay Module shall have drivers with appropriately sized suppressions
- The Relay Module shall have drivers with appropriately sized suppressions, such as a flyback diode
- The Relay Module shall have drivers with appropriately sized suppressions, such as a TVS/clamp network

2.1.1 The Relay Module shall provide eight indipendently controlled relay-driven channels

Statements

- The Relay Module shall provide eight indipendently controlled relay-driven channels
- The Relay Module shall manage its eight physical driver outputs

2.1.2 The Relay Module shall manage activation of each driver channel

Statements

- The Relay Module shall manage activation of each channel

2.1.3 The Relay Module shall manage deactivation of each driver channel

Statements

- The Relay Module shall manage deactivation of each channel

2.1.4 The Relay Module shall energize each relay coil independently across the full specified supply

Statements

- The Relay Module shall reliably energize its assigned relay coil across the full specified supply
- The Relay Module shall independently energize its assigned relay coil across the full specified supply

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Statements

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- The Relay Module shall energize its assigned relay coil across the full specified tolerance range

2.1.6 The Relay Module shall energize its assigned relay coil with adequate voltage margin

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- The Relay Module shall limit the coil-generated transient to a safe level without exceeding driver ratings

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- The Relay Module shall limit the coil-generated transient to a safe level without causing unacceptable relay release-time delays

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- The Relay Module shall have drivers with appropriately sized suppressions

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- The Relay Module shall have drivers with appropriately sized suppressions, such as a flyback diode

2.1.17 The Relay Module shall have drivers with appropriately sized suppressions, such as a TVS/clamp network

Statements

- The Relay Module shall have drivers with appropriately sized suppressions, such as a TVS/clamp network

2.2 Module Power Supply

- The Relay Module shall operate correctly over the complete specified nominal supply range
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- The Relay Module shall operate correctly over the complete specified nominal supply range, including steady-state variation
- The Relay Module shall operate correctly over the complete specified nominal supply range, including startup conditions
- The Relay Module shall operate correctly over the complete specified nominal supply range, including brownouts
- The Relay Module shall operate correctly over the complete specified nominal supply range, including overvoltage
- The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple
- The Relay Module shall operate correctly over the complete specified nominal supply range, including transient disturbances from cabinet
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- The Relay Module shall operate correctly over the complete specified nominal supply range, including component tolerances

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- The Relay Module shall operate correctly over the complete specified nominal supply range, including brownouts

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- The Relay Module shall operate correctly over the complete specified nominal supply range, including overvoltage

2.2.7 The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple

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- The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple

2.2.8 The Relay Module shall operate correctly over the complete specified nominal supply range, including transient disturbances from cabinet

Statements

- The Relay Module shall operate correctly over the complete specified nominal supply range, including transient disturbances from cabinet

2.2.9 The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple

Statements

- The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple

2.3 Module Startup

- The Relay Module shall include a startup sequence that verifies all required permissives
- The Relay Module shall include a startup sequence that establishes the required gas flow
- The Relay Module shall include a startup sequence that establishes the supporting utilities
- The Relay Module shall include a startup sequence that enables cooling
- The Relay Module shall include a startup sequence that enables other auxiliaries
- The Relay Module shall include a startup sequence that confirms healthy operating feedback of auxiliaries
- The Relay Module shall include a startup sequence that energizes the ozone-generation system
- The Relay Module shall include a startup sequence that transitions Aurora to the normal ozone-production state when the startup sequence is complete
- The Relay Module shall initialize deterministically with outputs disabled until valid control from the CORE is established after power restoration

2.3.1 The Relay Module shall include a startup sequence that verifies all required permissives

Statements

- The Relay Module shall include a startup sequence that verifies all required permissives

2.3.2 The Relay Module shall include a startup sequence that establishes the required gas flow

Statements

- The Relay Module shall include a startup sequence that establishes the required gas flow

2.3.3 The Relay Module shall include a startup sequence that establishes the supporting utilities

Statements

- The Relay Module shall include a startup sequence that establishes the supporting utilities

2.3.4 The Relay Module shall include a startup sequence that enables cooling

Statements

- The Relay Module shall include a startup sequence that enables cooling

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Statements

- The Relay Module shall include a startup sequence that enables other auxiliaries

2.3.6 The Relay Module shall include a startup sequence that confirms healthy operating feedback of auxiliaries

Statements

- The Relay Module shall include a startup sequence that confirms healthy operating feedback of auxiliaries

2.3.7 The Relay Module shall include a startup sequence that energizes the ozone-generation system

Statements

- The Relay Module shall include a startup sequence that energizes the ozone-generation system

2.3.8 The Relay Module shall include a startup sequence that transitions Aurora to the normal ozone-production state when the startup sequence is complete

Statements

- The Relay Module shall include a startup sequence that transitions Aurora to the normal ozone-production state when the startup sequence is complete

2.3.9 The Relay Module shall initialize deterministically with outputs disabled until valid control from the CORE is established after power restoration

Statements

- The Relay Module shall initialize deterministically with outputs disabled until valid control from the CORE is established after power restoration

2.4 Module Shutdown

- The Relay Module shall include a shutdown sequence that disables ozone generation
- The Relay Module shall include a shutdown sequence that maintains the required cooling
- The Relay Module shall include a shutdown sequence that maintains the required gas-flow/purge functions for the specified post-run period
- The Relay Module shall include a shutdown sequence that maintains the safely de-energizes the remaining auxiliaries in the defined order
- The Relay Module shall include a shutdown sequence that maintains the confirms that the process has reached the stopped state

2.4.1 The Relay Module shall include a shutdown sequence that disables ozone generation

Statements

- The Relay Module shall include a shutdown sequence that disables ozone generation

2.4.2 The Relay Module shall include a shutdown sequence that maintains the required cooling

Statements

- The Relay Module shall include a shutdown sequence that maintains the required cooling

2.4.3 The Relay Module shall include a shutdown sequence that maintains the required gas-flow/purge functions for the specified post-run period

Statements

- The Relay Module shall include a shutdown sequence that maintains the required gas-flow/purge functions for the specified post-run period

2.4.4 The Relay Module shall include a shutdown sequence that maintains the safely de-energizes the remaining auxiliaries in the defined order

Statements

- The Relay Module shall include a shutdown sequence that maintains the safely de-energizes the remaining auxiliaries in the defined order

2.4.5 The Relay Module shall include a shutdown sequence that maintains the confirms that the process has reached the stopped state

Statements

- The Relay Module shall include a shutdown sequence that maintains the confirms that the process has reached the stopped state

2.5 Module Permit

- The Relay Module shall permit ozone generation only when all required utilities are available
- The Relay Module shall permit ozone generation only when gas flow is within their specified limits
- The Relay Module shall permit ozone generation only when cooling is within their specified limits
- The Relay Module shall permit ozone generation only when required valves are in their correct states
- The Relay Module shall permit ozone generation only when required auxiliaries are in their correct states
- The Relay Module shall permit ozone generation only when no active safety interlock is violated
- The Relay Module shall permit ozone generation only when no active process interlock is violated
- The Relay Module shall permit ozone generation only when all required feedback signals are valid
- The Relay Module shall permit ozone generation only when the system is in an authorized operating mode

2.5.1 The Relay Module shall permit ozone generation only when all required utilities are available

Statements

- The Relay Module shall permit ozone generation only when all required utilities are available

2.5.2 The Relay Module shall permit ozone generation only when gas flow is within their specified limits

Statements

- The Relay Module shall permit ozone generation only when gas flow is within their specified limits

2.5.3 The Relay Module shall permit ozone generation only when cooling is within their specified limits

Statements

- The Relay Module shall permit ozone generation only when cooling is within their specified limits

2.5.4 The Relay Module shall permit ozone generation only when required valves are in their correct states

Statements

- The Relay Module shall permit ozone generation only when required valves are in their correct states

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Statements

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Statements

- The Relay Module shall permit ozone generation only when no active safety interlock is violated

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- The Relay Module shall permit ozone generation only when no active process interlock is violated

2.5.8 The Relay Module shall permit ozone generation only when all required feedback signals are valid

Statements

- The Relay Module shall permit ozone generation only when all required feedback signals are valid

2.5.9 The Relay Module shall permit ozone generation only when the system is in an authorized operating mode

Statements

- The Relay Module shall permit ozone generation only when the system is in an authorized operating mode

2.6 Module Stop

- The Relay Module shall immediately stop ozone generation whenever a condition capable of creating an unsafe or uncontrolled process state is detected
- The Relay Module shall immediately stop ozone generation when detected a loss of required gas flow
- The Relay Module shall immediately stop ozone generation when detected a loss of required cooling
- The Relay Module shall immediately stop ozone generation when detected a critical equipment failure
- The Relay Module shall immediately stop ozone generation when detected hazardous process-variable limits
- The Relay Module shall immediately stop ozone generation when detected loss of required safety permissives
- The Relay Module shall immediately stop ozone generation when detected emergency shutdown
- The Relay Module shall immediately stop ozone generation when detected defined critical interlock

2.6.1 The Relay Module shall immediately stop ozone generation whenever a condition capable of creating an unsafe or uncontrolled process state is detected

Statements

- The Relay Module shall immediately stop ozone generation whenever a condition capable of creating an unsafe or uncontrolled process state is detected

2.6.2 The Relay Module shall immediately stop ozone generation when detected a loss of required gas flow

Statements

- The Relay Module shall immediately stop ozone generation when detected a loss of required gas flow

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Statements

- The Relay Module shall immediately stop ozone generation when detected a loss of required cooling

2.6.4 The Relay Module shall immediately stop ozone generation when detected a critical equipment failure

Statements

- The Relay Module shall immediately stop ozone generation when detected a critical equipment failure

2.6.5 The Relay Module shall immediately stop ozone generation when detected hazardous process-variable limits

Statements

- The Relay Module shall immediately stop ozone generation when detected hazardous process-variable limits

2.6.6 The Relay Module shall immediately stop ozone generation when detected loss of required safety permissives

Statements

- The Relay Module shall immediately stop ozone generation when detected loss of required safety permissives

2.6.7 The Relay Module shall immediately stop ozone generation when detected emergency shutdown

Statements

- The Relay Module shall immediately stop ozone generation when detected emergency shutdown

2.6.8 The Relay Module shall immediately stop ozone generation when detected defined critical interlock

Statements

- The Relay Module shall immediately stop ozone generation when detected defined critical interlock

2.7 Module Enforce

- The Relay Module shall locally enforce fault-handling
- The Relay Module shall locally enforce the safety-related output constraints for its eight relay channels
- The Relay Module shall locally enforce the fail-safe behavior for its eight relay channels
- The Relay Module shall locally enforce output-level safety constraints
- The Relay Module shall locally enforce watchdog/communication-loss responses

2.7.1 The Relay Module shall locally enforce fault-handling

Statements

- The Relay Module shall enforce local fault-handling
- The Relay Module shall locally enforce internal fault handling

2.7.2 The Relay Module shall locally enforce the safety-related output constraints for its eight relay channels

Statements

- The Relay Module shall locally enforce the safety-related output constraints for its eight relay channels

2.7.3 The Relay Module shall locally enforce the fail-safe behavior for its eight relay channels

Statements

- The Relay Module shall locally enforce the fail-safe behavior for its eight relay channels

2.7.4 The Relay Module shall locally enforce output-level safety constraints

Statements

- The Relay Module shall locally enforce output-level safety constraints

2.7.5 The Relay Module shall locally enforce watchdog/communication-loss responses

Statements

- The Relay Module shall locally enforce watchdog/communication-loss responses

2.8 Module Diagnostic

- The Relay Module shall locally perform hardware diagnostics

2.8.1 The Relay Module shall locally perform hardware diagnostics

Statements

- The Relay Module shall perform local hardware diagnostics

2.9 Module Safety

- The Relay Module shall transition all relay outputs to defined safe state during out-of-range brownout
- The Relay Module shall transition all relay outputs to defined safe state during power loss
- The Relay Module shall define safe-states for all its outputs (based on its associated hazard analysis)
- The Relay Module shall define OFF/de-energized as the default safe-state unless a documented safety analysis requires a specific output to remain energized
- The Relay Module shall transition its outputs to safe states when system-level fault

2.9.1 The Relay Module shall transition all relay outputs to defined safe state during out-of-range brownout

Statements

- The Relay Module shall transition all relay outputs to defined safe state during out-of-range brownout

2.9.2 The Relay Module shall transition all relay outputs to defined safe state during power loss

Statements

- The Relay Module shall transition all relay outputs to defined safe state during power loss

2.9.3 The Relay Module shall define safe-states for all its outputs (based on its associated hazard analysis)

Statements

- The Relay Module shall have a formally defined safe state for all its outputs based on its associated hazard analysis

2.9.4 The Relay Module shall define OFF/de-energized as the default safe-state unless a documented safety analysis requires a specific output to remain energized

Statements

- The Relay Module shall have OFF/de-energized as the default safe state unless a documented safety analysis requires a specific output to remain energized

2.9.5 The Relay Module shall transition its outputs to safe states when system-level fault

Statements

- The Relay Module shall transition its outputs to safe states when system-level fault
- The Relay Module shall transition to defined safe state when in failure state

2.10 Module Fault

- The Relay Module shall monitor output faults
- The Relay Module shall maintain the output defined safe behavior under fault conditions
- The Relay Module shall prevent one faulty channel to affect other channels
- The Relay Module shall detect relevant hardware faults
- The Relay Module shall prevent reactivation until the reauthorization conditions are satisfied when it detects an internal fault capable of compromising safe operation

2.10.1 The Relay Module shall monitor output faults

Statements

- The Relay Module shall monitor output faults
- The Relay Module shall detect relevant output faults

2.10.2 The Relay Module shall maintain the output defined safe behavior under fault conditions

Statements

- The Relay Module shall maintain the output defined safe behavior under fault conditions
- The Relay Module shall prevent unintended activation when in failure state
- The Relay Module shall transition affected outputs to their predefined safe states independently of higher-level system decisions
- The Relay Module shall immediately prevent unsafe output activation when it detects an internal fault capable of compromising safe operation
- The Relay Module shall force all affected outputs to their defined safe states when it detects an internal fault capable of compromising safe operation
- The Relay Module shall latch the fault as required when it detects an internal fault capable of compromising safe operation
- The Relay Module shall prevent reactivation until the fault-clear conditions are satisfied when it detects an internal fault capable of compromising safe operation
- The Relay Module shall locally enforce defined safe-state behavior without relying solely on the CORE

2.10.3 The Relay Module shall prevent one faulty channel to affect other channels

Statements

- The Relay Module shall prevent one faulty channel to affect other channels

2.10.4 The Relay Module shall detect relevant hardware faults

Statements

- The Relay Module shall detect relevant hardware faults

2.10.5 The Relay Module shall prevent reactivation until the reauthorization conditions are satisfied when it detects an internal fault capable of compromising safe operation

Statements

- The Relay Module shall prevent reactivation until the reauthorization conditions are satisfied when it detects an internal fault capable of compromising safe operation

3. Relay

3.1 Relay Coil

- The Relay Module shall operate with coils that have a defined nominal voltage range
- The Relay Module shall operate with coils that have defined nominal current
- The Relay Module shall operate with coils that have defined maximum current
- The Relay Module shall operate with coils that have defined resistance
- The Relay Module shall operate with coils that have defined inductance
- The Relay Module shall operate with coils that have defined inrush behavior
- The Relay Module shall operate with coils that have defined continuous-duty rating
- The Relay Module shall operate with coils that have operate characteristics that are fully compatible with its corresponding driver
- The Relay Module shall operate with coils that have release characteristics that are fully compatible with its corresponding driver

3.1.1 The Relay Module shall operate with coils that have a defined nominal voltage range

Statements

- The Relay Module shall operate with coils that have a defined nominal voltage
- The Relay Module shall operate with coils that have allowable voltage range

3.1.2 The Relay Module shall operate with coils that have defined nominal current

Statements

- The Relay Module shall operate with coils that have nominal current

3.1.3 The Relay Module shall operate with coils that have defined maximum current

Statements

- The Relay Module shall operate with coils that have maximum current

3.1.4 The Relay Module shall operate with coils that have defined resistance

Statements

- The Relay Module shall operate with coils that have defined resistance

3.1.5 The Relay Module shall operate with coils that have defined inductance

Statements

- The Relay Module shall operate with coils that have defined inductance

3.1.6 The Relay Module shall operate with coils that have defined inrush behavior

Statements

- The Relay Module shall operate with coils that have defined inrush behavior

3.1.7 The Relay Module shall operate with coils that have defined continuous-duty rating

Statements

- The Relay Module shall operate with coils that have defined continuous-duty rating

3.1.8 The Relay Module shall operate with coils that have operate characteristics that are fully compatible with its corresponding driver

Statements

- The Relay Module shall operate with coils that have operate characteristics that are fully compatible with its corresponding driver

3.1.9 The Relay Module shall operate with coils that have release characteristics that are fully compatible with its corresponding driver

Statements

- The Relay Module shall operate with coils that have release characteristics that are fully compatible with its corresponding driver